Pingyue Zhang

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Research Interest

Spatial cognition in foundation models, from static spatial reasoning to interactive exploration and robotics

Education

Ph.D., Computer Science, *Northwestern University, IL, USA* *Sept 2024 - present*

- Supervisor: Prof. Manling Li

M.Eng., Computer Science, *Shanghai Jiao Tong University, Shanghai, China* *Sept 2021 - Mar 2024*

- Grades (Cumulative GPA): 3.88/4.0
- Supervisor: Prof. Mengyue Wu

B.Eng. (Honor), Computer Science (IEEE Honor Class), *Shanghai Jiao Tong University, Shanghai, China* *Sept 2017 - June 2021*

- Grades (Cumulative Scores): 90/100, Ranking: 11/84

Honor & Awards

Research Awards

- Best Poster Award, Midwest Machine Learning Symposium (MMLS) 2025, for RAGEN
- Spotlight, ACL 2026 Workshop on Advances in Language and Vision Research, for Theory of Space
- Spotlight, ICCV 2025 Workshop on Embodied Spatial Reasoning, for MindCube

Scholarship

- Zhiyuan College Honors Scholarship (top 5%, RMB 5000 per year, 2017 - 2021)
- HYPERGRYPH Scholarship (RMB 5000, Sept 2022)

Travel Grant

- SIGMM Travel Grants for ACM MM 2021 ($1000)

Competition

- Silver medal in IBM Call For Code Competition, Shanghai (RMB 10000, July 2019)

Research Experience

PhD at Northwestern University *(Advised by Prof. Manling Li)* *Sept 2024 - Present*

- **Spatial cognition in foundation models: from static reasoning to interaction and robotics.**
  - **MindCube** (ICLR 2026): spatial mental modeling from limited views; map-then-reason training raises accuracy from 37.8% to 70.7%.
  - **Theory of Space** (first author, ICLR 2026): benchmark and belief-probing method for active spatial exploration; identified the active-passive gap and belief inertia in foundation models.
  - **Embodied First Exam** (co-first author, 2026): helped design the benchmark and implemented its framework and tasks; 1,108 robotics tasks from 33 suites, 46 robots and 22 simulators for code agents; the best agent solves 76% of tasks with simulator state but 28% with robot sensors only.
  - **RAGEN** and **VAGEN** (co-first author; NeurIPS 2025): open-source multi-turn RL frameworks for LLM and VLM agents (RAGEN: 2,800+ GitHub stars).
  - Ongoing: SpatialGym, an interactive benchmark of atomic spatial skills, and latent spatio-temporal (4D) reasoning in VLMs.

Research Intern, Adobe Research *June 2026 - Sept 2026*

- Spatio-temporal (4D) reasoning in vision-language models.

Research in Shanghai Jiao Tong University X-LANCE LAB *(Directed by Prof. Mengyue Wu)* *Sept 2019 - Mar 2024*

- **Automatic detection of disease like depression or Parkinson based on audio or text data.**
- **Enhance the contrastive learning method to improve classification performance.**
- **Zero-shot learning.**

Publications

Embodied First Exam.

Kangrui Wang, Qineng Wang*, Changhe Chen*, **Pingyue Zhang***, Chi Gui*, Feishi Wang, Canyu Chen, Wenli Xiao, Xiaoyu Huang, Tonghe Zhang, Linjie Li, Zihan Wang, Minyeong Kim, Yiyou Sun, Mingkai Deng, Jinyu Hou, Eric P. Xing, Han Liu, Qi Zhu, Huaxiu Yao, Zhiwen Fan, Yuejiang Liu, Dawn Song, Pieter Abbeel, Jiajun Wu, Yuke Zhu, Guanya Shi, Ruohan Zhang, Manling Li.*

2026 [Website]

Theory of Space: Can Foundation Models Construct Spatial Beliefs through Active Exploration?
Pingyue Zhang*, *Zihan Huang*, Yue Wang*, Jieyu Zhang*, Letian Xue, Zihan Wang, Qineng Wang, Keshigeyan Chandrasegaran, Ruohan Zhang, Yejin Choi, Ranjay Krishna, Jiajun Wu, Li Fei-Fei, Manling Li.*
Spotlight at ACL workshop on Advances in Language and Vision Research, 2026
ICLR, 2026

RAGEN: Understanding Self-Evolution in LLM Agents via Multi-Turn Reinforcement Learning.
Zihan Wang, Kangrui Wang*, Qineng Wang*, **Pingyue Zhang*,** Linjie Li*, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, Eli Gottlieb, Yiping Lu, Kyunghyun Cho, Jiajun Wu, Li Fei-Fei, Lijuan Wang, Yejin Choi, Manling Li.* arXiv, 2025
Best Poster Award at MMLS 2025 (Midwest Machine Learning Symposium)

MindCube: Spatial Mental Modeling from Limited Views.
Baiqiao Yin, Qineng Wang*, **Pingyue Zhang,** Jianshu Zhang, Kangrui Wang, Zihan Wang, Jieyu Zhang, Keshigeyan Chandrasegaran, Han Liu, Ranjay Krishna, Saining Xie, Manling Li, Jiajun Wu, Li Fei-Fei.*
Spotlight at ICCV Workshop on Embodied Spatial Reasoning
ICLR, 2026

VAGEN: Reinforcing World Model Reasoning for Multi-Turn VLM Agents.
Kangrui Wang, **Pingyue Zhang*,** Zihan Wang*, Yaning Gao*, Linjie Li*, Qineng Wang, Hanyang Chen, Chi Wan, Yiping Lu, Zhengyuan Yang, Lijuan Wang, Ranjay Krishna, Jiajun Wu, Li Fei-Fei, Yejin Choi, Manling Li.*
NeurIPS, 2025

RAGEN-2: Reasoning Collapse in Agentic RL.
*Zihan Wang, Chi Gui, Xing Jin, Qineng Wang, Licheng Liu, Kangrui Wang, Shiqi Chen, Linjie Li, Zhengyuan Yang, **Pingyue Zhang**, Yiping Lu, Jiajun Wu, Li Fei-Fei, Lijuan Wang, Yejin Choi, Manling Li.*
ICML, 2026 (Oral)

WorldAgen: Unified State-Action Prediction with Test-Time World Model Training.
*Chi Wan, Kangrui Wang, Yuan Si, **Pingyue Zhang**, Manling Li.*
AAAI, 2026

CURV: Enhancing Chart Understanding Through Curriculum Visual Grounded Reasoning.
*Xuehang Guo, **Pingyue Zhang**, Ruiyi Zhang, Zhenhailong Wang, Hanrui Lyu, Heng Ji, Tong Sun, Qingyun Wang, Manling Li.*
arXiv, 2026

LEMONADE: A Large Multilingual Expert-Annotated Abstractive Event Dataset for the Real World.
*Sina J Semnani, **Pingyue Zhang,** Wanyue Zhai, Haozhuo Li, Ryan Beauchamp, Trey Billing, Katayoun Kishi, Manling Li, Monica S Lam.*
(ACL 2025 Findings)

Multi-Label Supervised Contrastive Learning.
Zhang, P., *& Wu, M. (2024).* (AAAI 2024)

Depa: Self-supervised audio embedding for depression detection. In Proceedings of the 29th ACM international conference on multimedia (pp. 135-143).
Zhang, P., *Wu, M., Dinkel, H., & Yu, K. (2021, October).* (ACMMM 2021)

Other publications from SJTU (audio and representation learning).
Semantic-Enhanced Supervised Contrastive Learning (first author, ICASSP 2024, oral); Enhancing Zero-shot Audio Classification using Sound Attribute Knowledge from LLMs (co-first author, Interspeech 2024); ReCLR: Reference-Enhanced Contrastive Learning of Audio Representation for Depression Detection (first author, Interspeech 2023, oral); BLAT: Bootstrapping Language-Audio Pre-training (ACM MM 2023).

Open-source projects

- RAGEN: multi-turn reinforcement learning for LLM agents (2,800+ GitHub stars, Oct 2026). [GitHub]
- VAGEN: multi-turn reinforcement learning for VLM agents (500+ GitHub stars, Oct 2026). [GitHub]
- Embodied First Exam: open exam and leaderboard for code agents in robotics. [Website]
- Theory of Space: benchmark code and data. [GitHub]